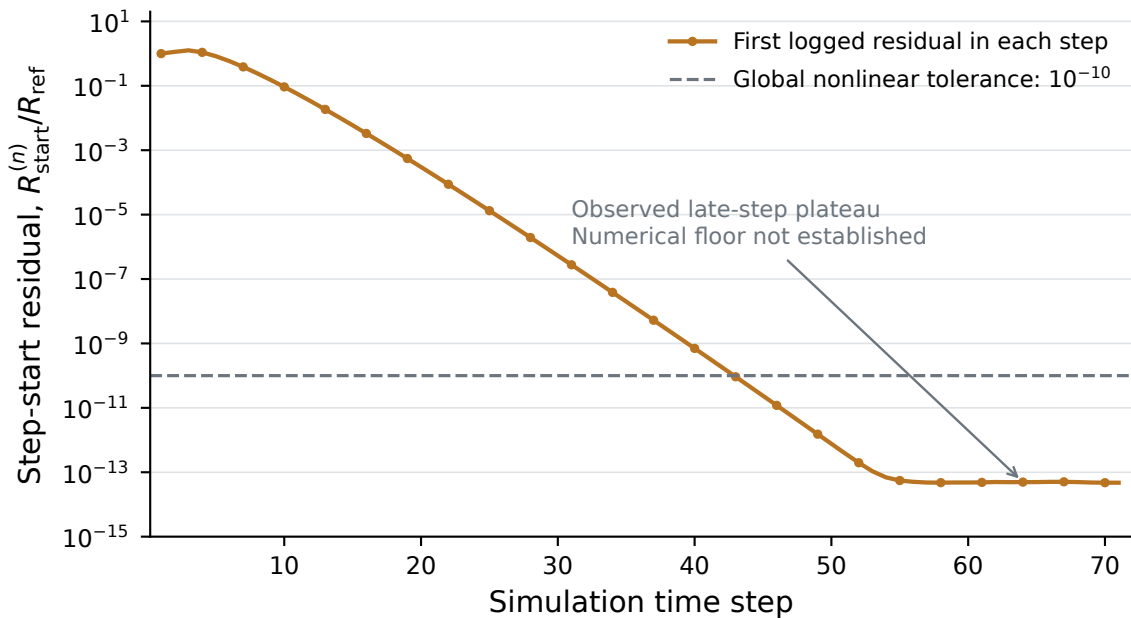


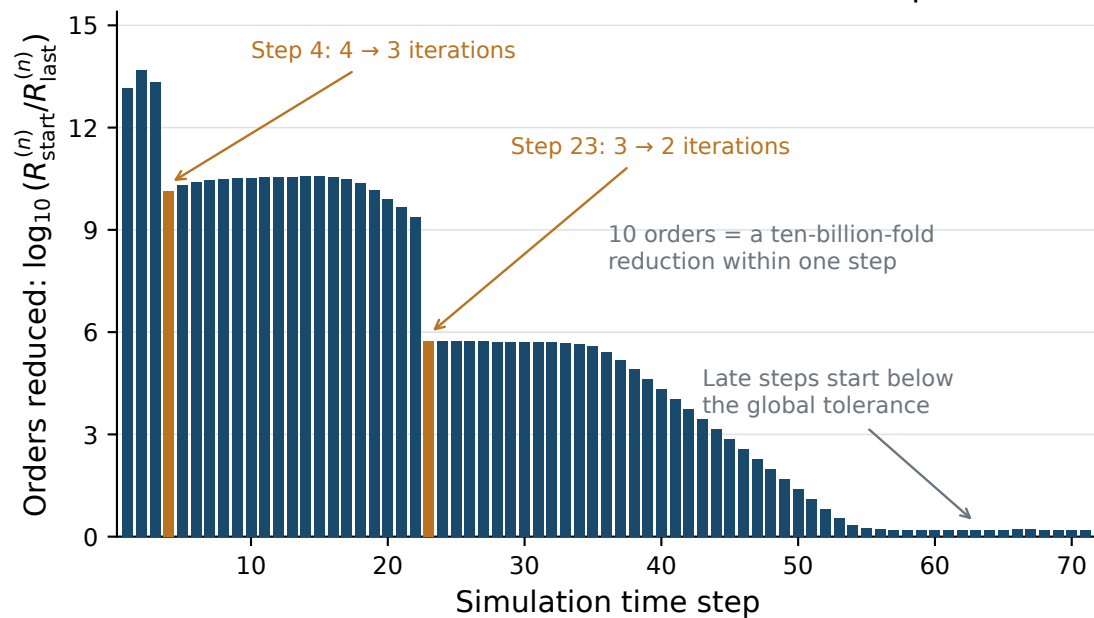
Solver Residual Diagnostics: Nonlinear and Linear Convergence

Nonlinear reference: $R_{\text{ref}} = R^{(1,1)}$, fixed for all 71 steps. Within-step reduction is evaluated separately.

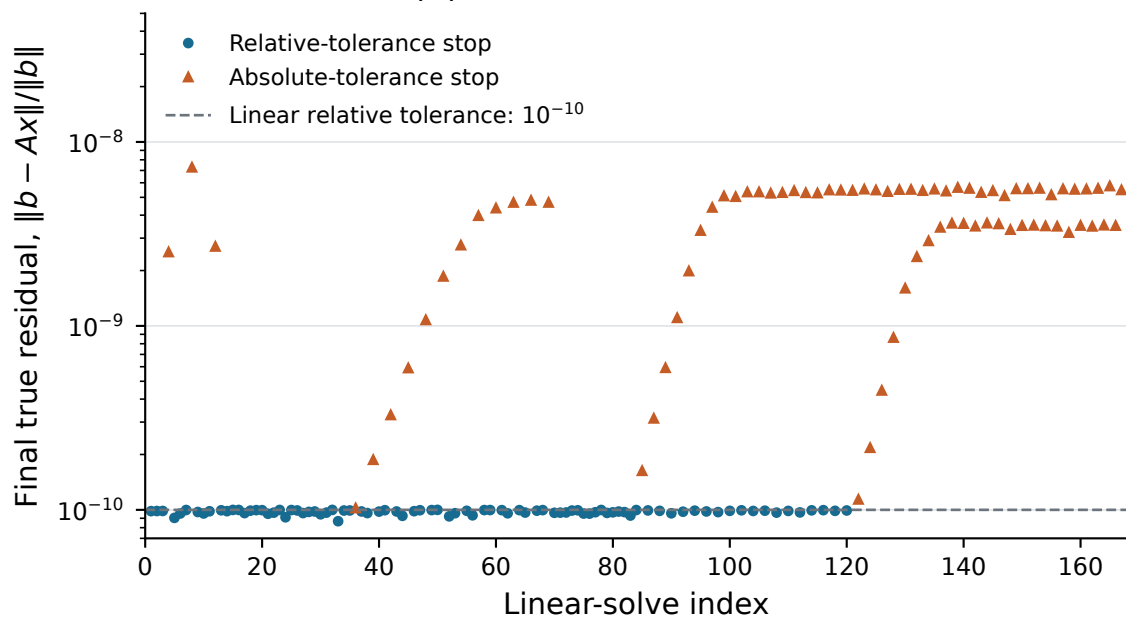
(A) Global transient evolution



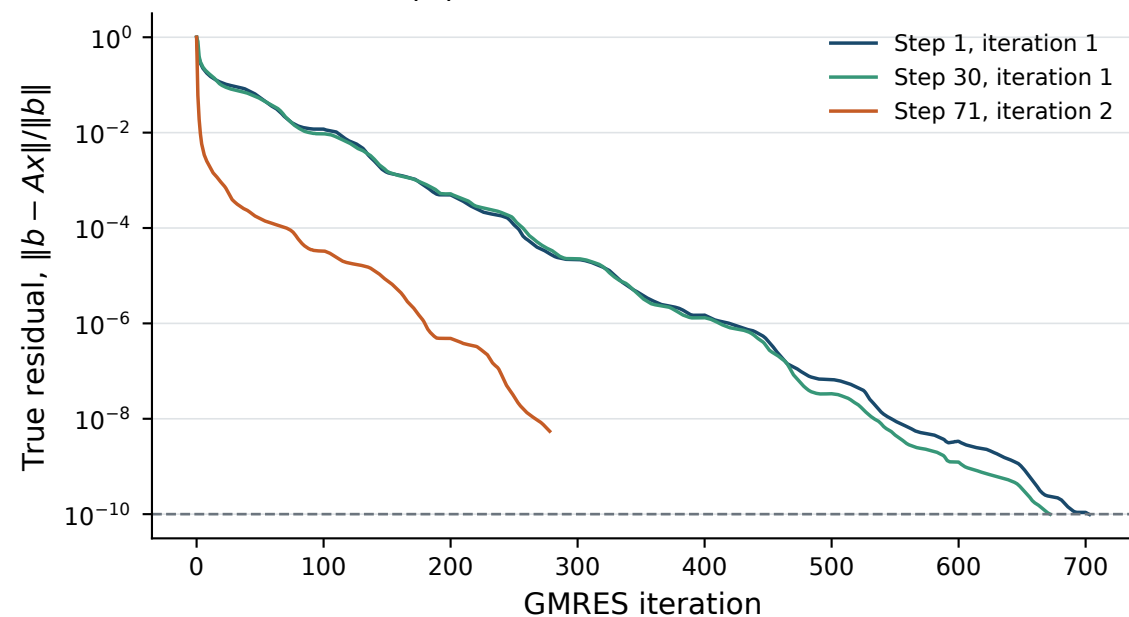
(B) Nonlinear reduction within each step



(C) Linear solve termination



(D) Selected GMRES histories



Nonlinear stop: at least 2 iterations and global OR step-relative ratio $\leq 10^{-10}$. Linear stop: relative 10^{-10} OR absolute 10^{-24} .

Residuals assess the discretized algebraic solve; they do not establish CFD solution accuracy or a floating-point floor.